MID-SEMESTER EXAMINATION

Subject: Data Structures, Algorithms, yamini@gmail.com

Paper: Set A | Duration: 3 Hours | Max Marks: 100 | Date: 10-03-2026

Total Questions: 10 | Answer all questions.

Section 1: MCQ (4 Questions - 8 Marks)

Q1. [2 marks] (Easy)

Which of the following data structures operates on a Last-In, First-Out (LIFO) principle?

Options:
A) Queue B) Stack C) Linked List D) Tree

Q2. [2 marks] (Easy)

What is the worst-case time complexity of a linear search algorithm on an array of 'n' elements?

Options:
A) O(1) B) O(log n) C) O(n) D) O(n^2)

Q4. [2 marks] (Easy)

Which data structure is best suited for implementing a "undo" functionality in a text editor?

Options:
A) Queue B) Array C) Stack D) Hash Table

Q8. [2 marks] (Medium)

In a complete binary tree with 'N' nodes, what is the maximum possible height (number of edges from root to the deepest leaf)?

Options:
A) N B) log2(N) C) N/2 D) N*log2(N)

Section 2: Short Answer (6 Questions - 30 Marks)

Q3. [5 marks] (Easy)

How many characters (including special characters) are present in the string "yamini@gmail.com"?

Q5. [5 marks] (Medium)

Explain the primary difference in memory allocation between an array and a linked list.

Q6. [5 marks] (Medium)

Describe the main steps involved in searching for an element in a sorted array using Binary Search.

Q7. [5 marks] (Medium)

If "yamini@gmail.com" needs to be stored in a hash table, propose a simple hash function for this string.

Q9. [5 marks] (Hard)

You need to store multiple email addresses like "yamini@gmail.com" and efficiently retrieve all email addresses belonging to a specific domain (e.g., all "@gmail.com" addresses). Propose a data structure design that optimizes this particular retrieval operation.

Q10. [5 marks] (Hard)

Analyze the worst-case time complexity of QuickSort when the pivot is always chosen as the first element, and the input array is already sorted in ascending order. Justify your answer.

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